Topic 2 Complex Numbers

- 2.1 Defining Complex Numbers 80
- 2.2 Complex Arithmetic 90
- 2.3 Complex Numbers in Polar Form 112
- 2.4 Sketching Regions in the Complex Plane 146
- 2.5 Powers and Roots of Complex Numbers 158
- 2.6 Complex Representation of Trigonometric Functions 174
- 2.7 Factoring Polynomials 179

2.1 Defining complex numbers

Consider the equality $x^2 + 1 = 0$. Naively applying the quadratic formula we find:

$$x = \pm\sqrt{-1},$$

which is absurd.

But what if it was not?

2.1 Defining complex numbers: the number i

Complex numbers are a number system in which previously unsolvable polynomial equations have solutions. Complex numbers are used extensively in physics and engineering, to study electromagnetic waves and electric circuits for example.

The key idea is to introduce a new number i (sometimes called an imaginary number) with the property

$$i^2 = -1.$$

There is no real number with this property, and so $i \notin \mathbb{R}$. Since $i \notin \mathbb{R}$ we need to think carefully about the meaning of numbers like

$$3 + i \qquad \text{and} \qquad 4i$$

2.1 Defining complex numbers

Back to our example: $x^2 + 1 = 0$.
Applying the quadratic formula yields:

$$x = \pm\sqrt{-1},$$

Since $i^2 = -1$, we find $x = \pm i$.

2.1 Defining complex numbers

Example 2.1

Assuming the usual rules of arithmetic apply, find two solutions of

$$x^2 + 4 = 0.$$

2.1 Defining Complex Numbers

Example 2.2

By completing the square, find the solutions of

$$x^2 - 2x + 5 = 0.$$

In order to solve some equations, it seems we need numbers of the form $x + iy$ where $x, y \in \mathbb{R}$. This turns out to be enough to solve all possible polynomial equations.

2.1 Defining Complex Numbers

Definition 2.3

A complex number is a quantity consisting of a real number added to a real multiple of i. That is, a complex number is a number of the form.

$$z = x + iy$$

where $x, y \in \mathbb{R}$. The set of all complex numbers is denoted $\mathbb{C}$. That is,

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}.$$

2.1 Defining Complex Numbers

Example 2.4

1. Using the number i, write down two square roots of -25.
2. Simplify i^7.

2.1 Defining Complex Numbers: Real and Imaginary parts

A complex number z of the form $z = a + bi$ has two parts.

Definition 2.5

Let $z = x + iy \in \mathbb{C}$.

- x is called the real part of z and is denoted $\mathrm{Re}(z)$;
- y is called the imaginary part of z and is denoted $\mathrm{Im}(z)$.

Note that both $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ are real numbers.

Example 2.6

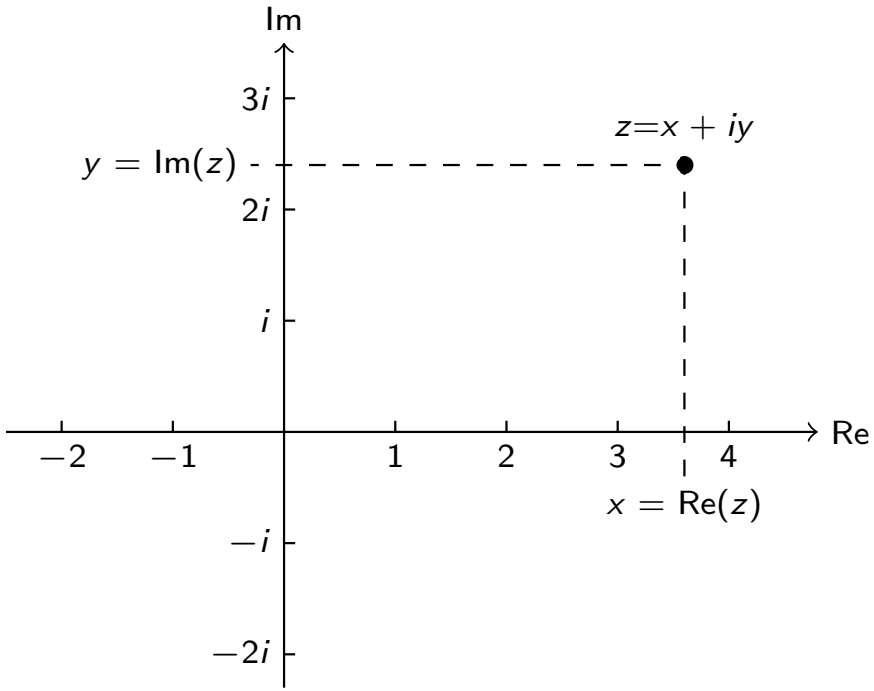
For the complex number $z = 3 - 4i$, write down:

1. $\mathrm{Re}(z)$.
2. $\mathrm{Im}(z)$.
3. $\sqrt{(\mathrm{Re}(z))^2 + (\mathrm{Im}(z))^2}$.

2.1 Complex Numbers: The Complex Plane

In our final computation, the computation we did felt a lot like finding the the length of a hypotenuse of a right triangle.

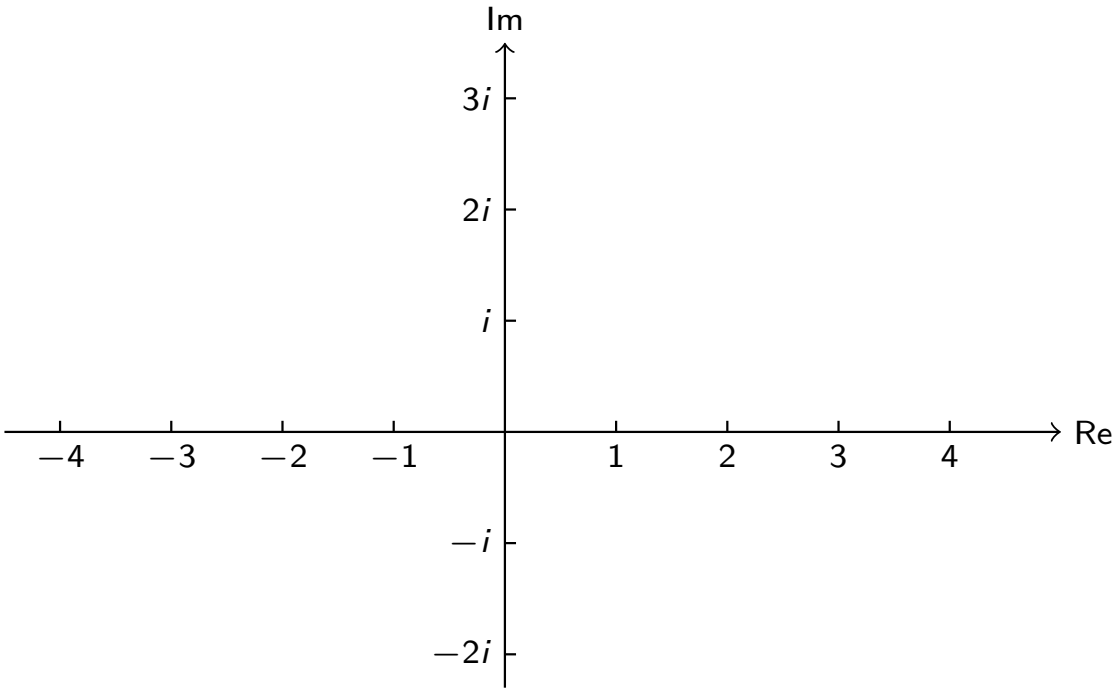
- Complex numbers can be represented graphically in the complex plane.
- A complex number $z = x + iy$ corresponds to a point in a plane with:
  - real part measured on the horizontal axis (the real axis)
  - imaginary part measured on the vertical axis (the imaginary axis).



2.1 Defining Complex Numbers: The Complex Plane

Example 2.7

Sketch the complex numbers $3 - i$, $-2 + 2i$, -4, and $3i$ in the complex plane.



2.2 Complex Arithmetic

Definition 2.8

Let $z_1 = a + ib$ and $z_2 = c + id$ be complex numbers $(a, b, c, d \in \mathbb{R})$. We say <u>$z_1$ and z_2 are equal</u> when $a = c$ and $b = d$.

We define operations on complex numbers as follows.

- <u>addition:</u> We add z_1 and z_2 as follows:

$$z_1 + z_2 = (a + ib) + (c + id) = (a + c) + i(b + d).$$

- <u>subtraction:</u> We subtract z_2 from z_1 as follows:

$$z_1 - z_2 = (a + ib) - (c + id) = (a - c) + i(b - d).$$

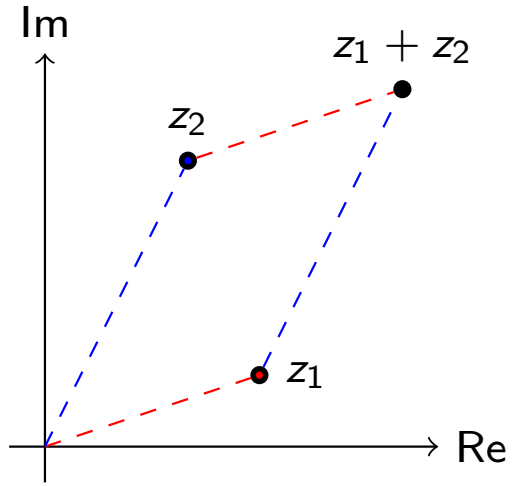
- <u>multiplication by a real</u>: We multiply z_1 by $k \in \mathbb{R}$ as follows:

$$kz_1 = k(a + ib) = (ka) + i(kb).$$

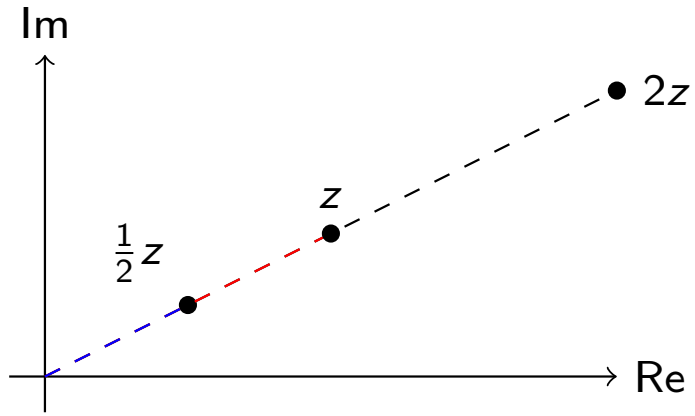
2.2 Complex Arithmetic

Each of our operations has an interpretation in the complex plane.

- $z_1 + z_2$ is the other vertex of the parallelogram with sides 0 to z_1 and 0 to z_2.



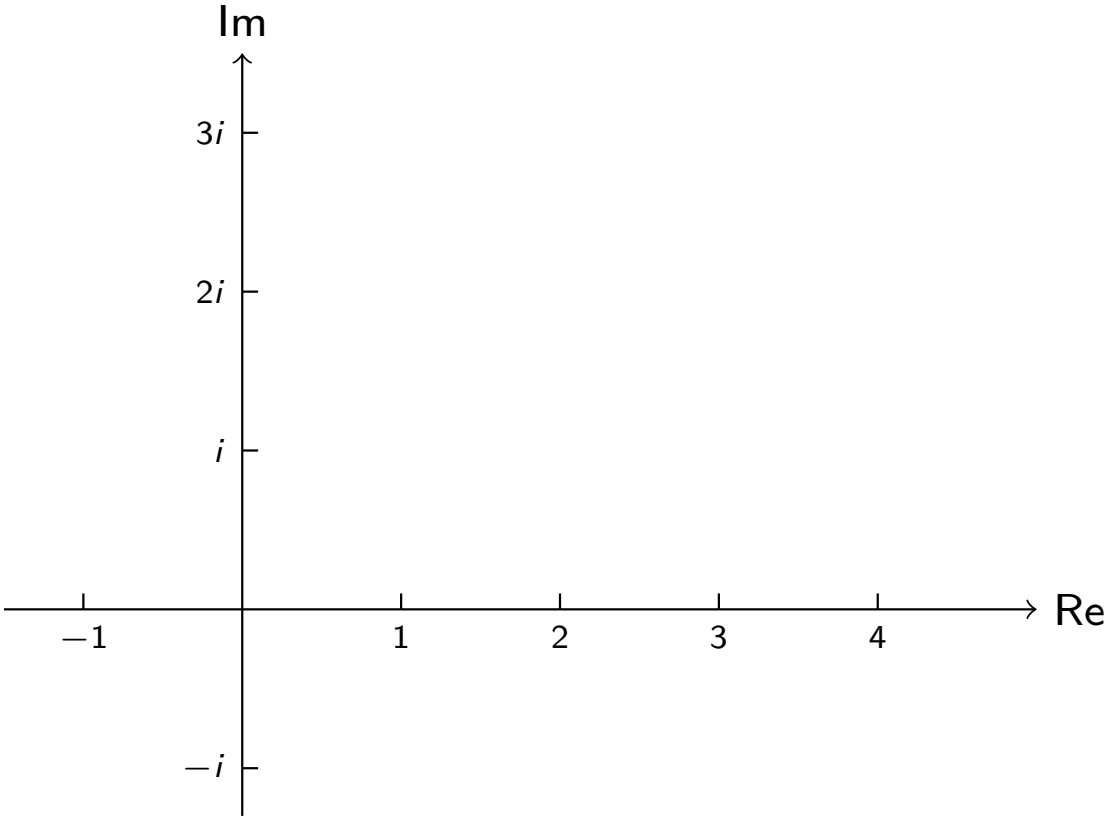
- For k positive, kz is found by stretching (or shrinking) z.



2.2 Complex Arithmetic

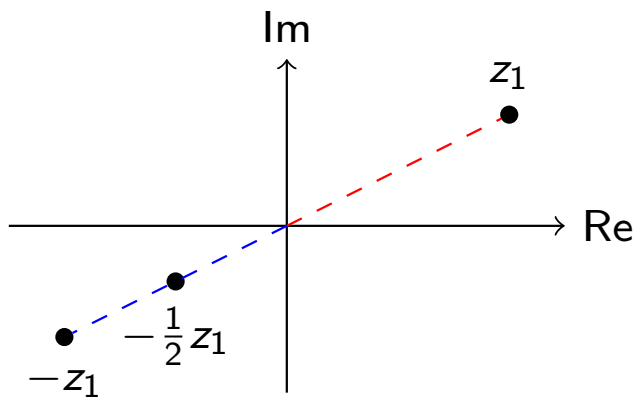
Example 2.9

Illustrate the geometric effect of adding complex numbers for $z_1 = 1 + 2i$ and $z_2 = 2 + i$.

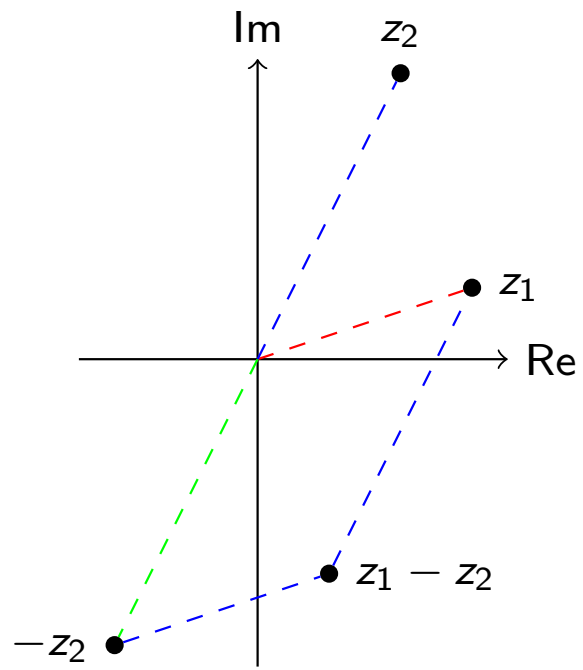


2.2 Complex Arithmetic

- For k negative, kz_1 is found by stretching or shrinking z_1 and reflecting through 0.
  In particular, $-z$ is a reflection through 0.



- $z_1 - z_2$ is similarly found by expressing it as $z_1 + (-z_2)$.



- We will explore these ideas further when we study vectors.

2.2 Complex Arithmetic: Complex Multiplication

- For convenience, it would be nice to have complex numbers obey the usual laws of arithmetic.
- Accordingly, we define multiplication using the familiar distributive law.

$$(a + ib)(c + id) =$$

2.2 Complex Arithmetic: Complex Division

We have defined addition, subtraction and multiplication of complex numbers. It seems reasonable that next would be complex division.
This takes a little more care than we might first expect.

Consider the following complex quotient

$$\frac{1+2i}{-1+3i}$$

2.2 Complex Arithmetic: Complex Division

Let $z = \frac{1+2i}{-1+3i}$. Simplifying here requires a bit of non-obvious cleverness:

2.2 Complex Arithmetic: the Complex Conjugate

In our simplification of a rational complex number, multiplying the denominator of $-1+3i$ by $-1-3i$ resulted in a real denominator.

Definition 2.10

Let $z = a + ib$ be a complex number. The <u>complex conjugate of z</u> is $a - ib$ and is denoted $\bar{z}$. That is, $\bar{z} = a - ib$.

2.2 Complex Arithmetic: the Complex Conjugate

Example 2.11

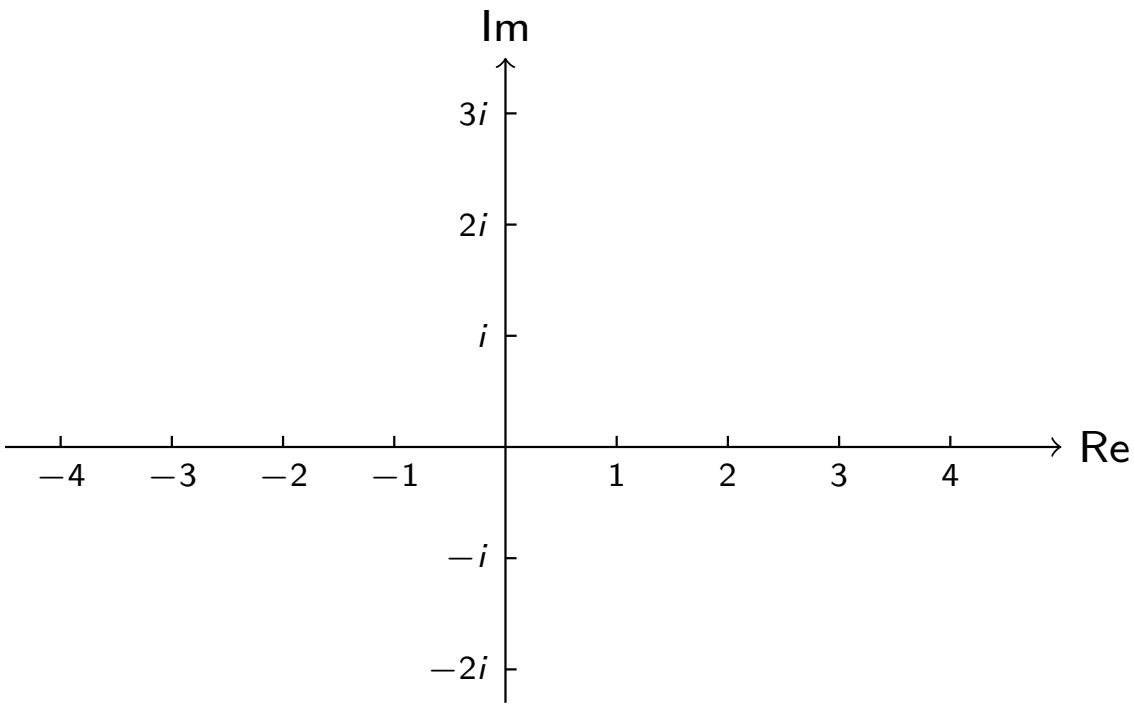
Write down the complex conjugates of:

(a) $-3 + 7i$ (b) $2 - 5i$ (c) $3i$ (d) 4

2.2 Complex Arithmetic: the Complex Conjugate

Example 2.12

- If $z_1 = 1 + i$ and $z_2 = -3 - 2i$, plot z_1, z_2, $\bar{z}_1$ and $\bar{z}_2$ in the complex plane.
- What is the geometric relationship between z and $\bar{z}$?



2.2 Complex Arithmetic: the Complex Conjugate

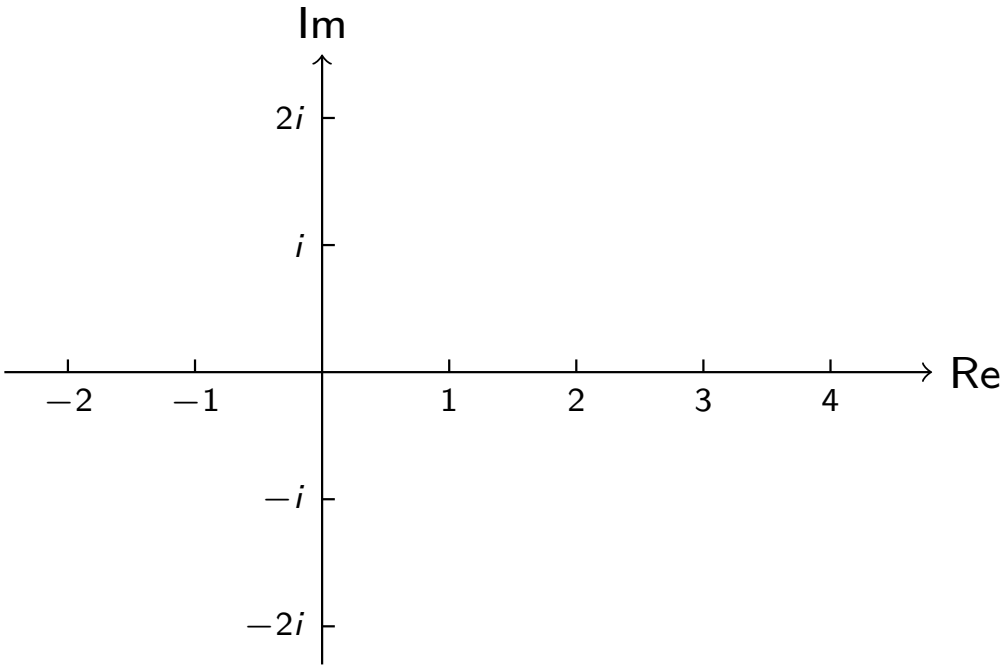
Homework 8

Plot $z_1 = 2i$, $z_2 = 3$, $\bar{z}_1$ *and* $\bar{z}_2$ *in the complex plane.*

2.2 Complex Arithmetic: the Complex Conjugate

Example 2.13

By completing the square, find the solutions of $z^2 - 6z + 10 = 0$ and plot them in the complex plane.



2.2 Complex Arithmetic: the Complex Conjugate

2.2 Complex Arithmetic: the Complex Conjugate

Theorem 2.14 (Properties of complex conjugates)

Let $z = x + iy$ and $w = a + ib$ be complex numbers. Then:

1. $z + \bar{z} = 2x = 2\,\mathrm{Re}(z)$
2. $z - \bar{z} = 2yi = 2\,\mathrm{Im}(z)i$
3. $z\bar{z} = x^2 + y^2$
4. $\overline{z + w} = \bar{z} + \bar{w}$
5. $\overline{zw} = \bar{z}\bar{w}$

We prove parts 1, 2 and 3 of Theorem 2.14 by applying the definition of complex conjugate, the meaning of the notation $\bar{z}$, and the meaning of the notation $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$

2.2 Complex Arithmetic: the Complex Conjugate

2.2 Complex Arithmetic: the Complex Conjugate

Homework 9

Prove part 4 of Theorem 2.14 for a pair of complex numbers $z = x + iy$ *and* $w = a + ib$.

2.2 Complex Arithmetic: the Complex Conjugate

Example 2.15

Using the complex conjugate, express simplify the following to the form $x + yi$.

$$z = \frac{1 + 2i}{-1 + 3i}$$

2.2 Complex Arithmetic

Example 2.16

Find $\operatorname{Re}\left(\dfrac{1+5i}{2-2i}\right)$ and $\operatorname{Im}\left(\dfrac{1+5i}{2-2i}\right)$.

2.2 Complex Arithmetic

Homework 10

Let $z = 1 - 5i$ *and* $w = -2 + i$*. Express the following complex numbers in the form* $a + ib$

(a) $w^2 z$

(b) $\dfrac{w}{w + 2z}$

2.2 Complex Arithmetic: Complex Exponentiation

Example 2.17

Calculate

$$(1+\sqrt{3}i)^6.$$

2.2 Complex Arithmetic: Complex Exponentiation

Example 2.18

Calculate

$$(1+\sqrt{3}i)^{6000}.$$

2.3 Polar Form

We have learned how to work with complex numbers written in <u>Cartesian form</u>

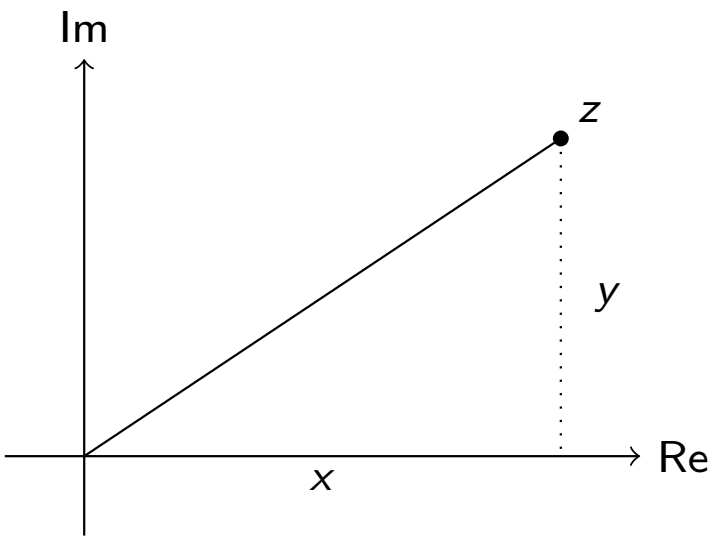
$$x + iy.$$

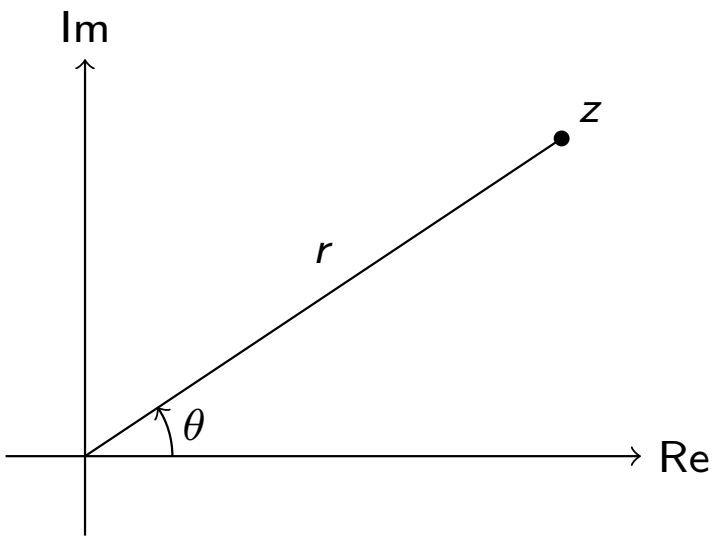
In this form, the task of adding and subtracting complex numbers is easy. Multiplication and division are reasonably easy in Cartesian form, but taking powers (and roots) is generally hard work. Our practice of representing complex numbers as points in the plane leads to another representation.

Instead of giving the x and y coordinates, we can represent a complex number z by saying how far it is from the origin and what angle we get when we draw a line from z to the origin. In this polar form finding powers and roots becomes much easier.

2.3 Polar Form: Modulus and Argument

- The position of a complex number z in the complex plane can be specified in two different ways:
  - Cartesian form: give the real and imaginary parts x and y, so that $z = x + iy$;
  - Polar form: give the distance r from the origin, and the angle θ from the positive real axis.





Note: Distance r is always given as a non-negative real.

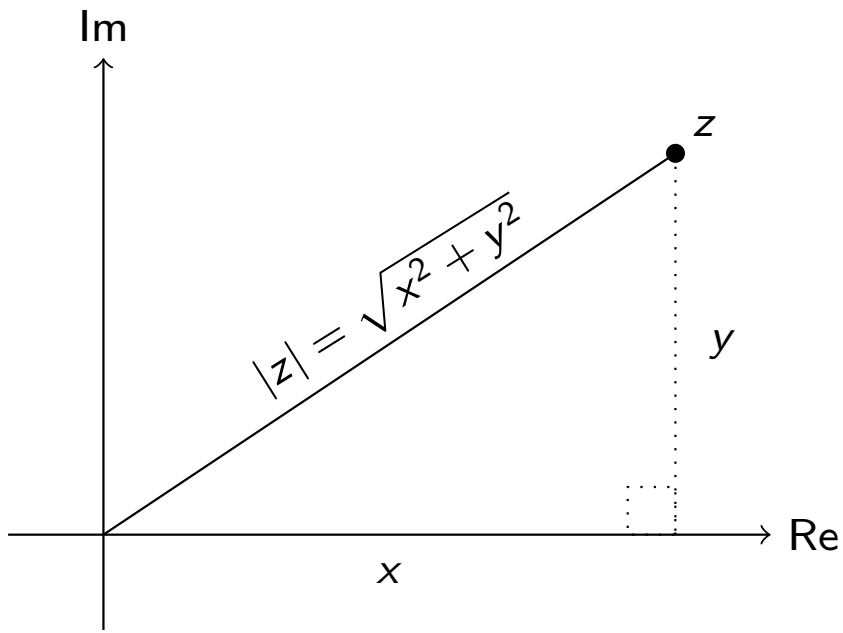
2.3 Polar Form: Modulus

Definition 2.19

Let $z \in \mathbb{C}$. The <u>modulus</u> of z is the distance r from z to the origin in the complex plane. We denote the modulus of z as $|z|$.

- Writing $z = x + iy$, we can find $|z|$ by Pythagoras' Theorem:

$$|z| = \sqrt{x^2 + y^2}.$$



2.3 Polar Form: Argument

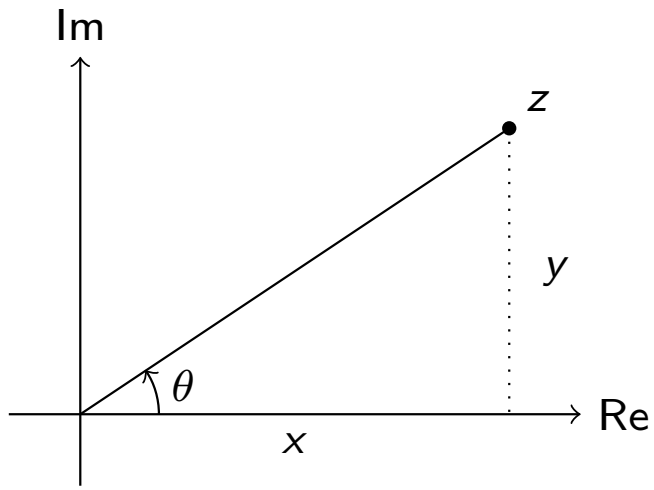
Definition 2.20

An <u>argument</u> of $z \neq 0$ is any angle θ that z makes with the positive real axis in the complex plane.

- To find an argument θ of

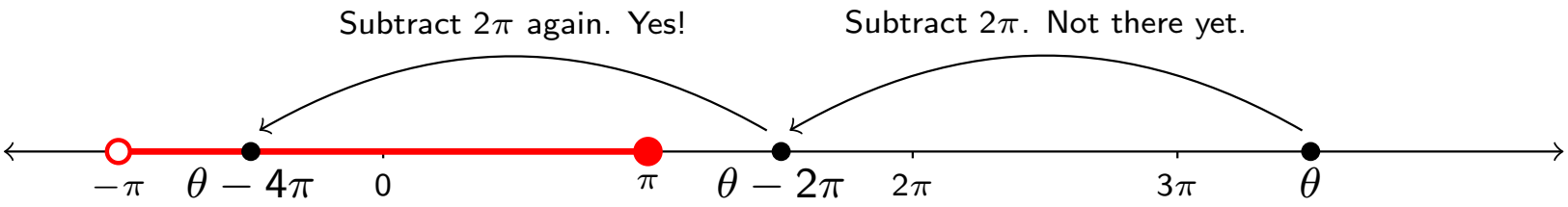
$$z = x + iy \neq 0$$

draw z in the complex plane and use standard triangles (if possible) to find θ.



2.3 Polar Form: Principal Argument

- Arguments of z are not unique. Adding multiples of 2π does not change the position of z in the complex plane.
- However, there is one unique argument in $(-\pi, \pi]$. We can find this argument by adding or subtracting multiples of 2π to any argument θ until we arrive at one in the interval $(-\pi, \pi]$.



- This is called the <u>principal argument</u> of z and is denoted $\mathrm{Arg}(z)$. Note the capital A.
- This gives a standard way of representing the angle - a bit like putting a fraction into the lowest terms:

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8} = \frac{5}{10} = \dots$$
$$\frac{\pi}{4} \approx \frac{9\pi}{4} \approx \frac{17\pi}{4} \approx \frac{25\pi}{4} \approx \dots$$

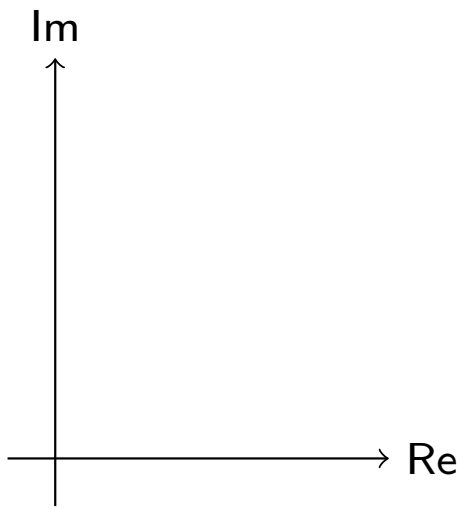
2.3 Polar Form: Modulus and Argument

Example 2.21

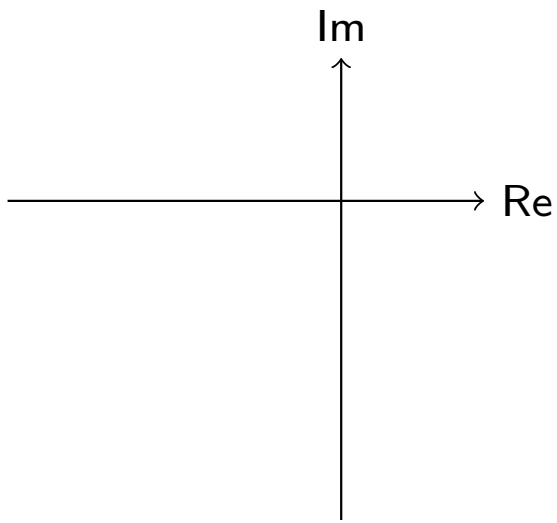
Find the modulus and an argument θ of

(a) $1 + \sqrt{3}i$ (b) $-3 - 3i$ (c) $3 + 4i$ (d) -1

(a)

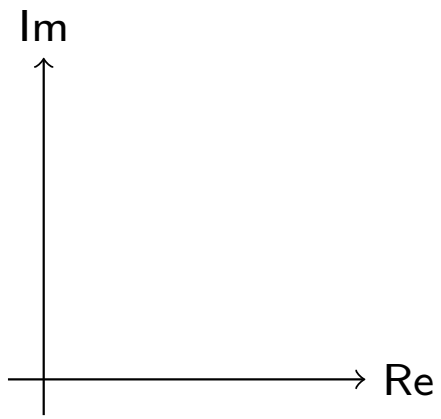


(b)

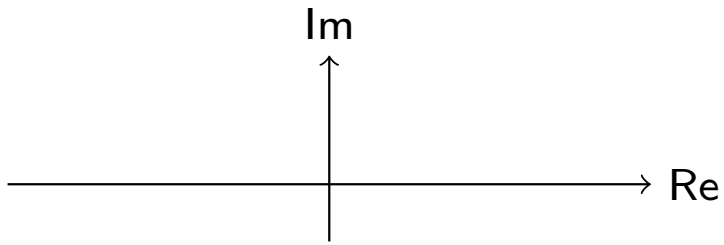


2.3 Polar Form: Modulus and Argument

(c)



(d)



- Always draw a sketch when converting to polar form!

2.3 Polar Form: Modulus and Argument

Homework 11

Find the modulus and an argument of $z = 2 - 2i$.

2.3 Polar Form: Modulus and Argument

- We can prove the following properties of the modulus and argument of complex numbers z and w.

Theorem 2.22 (Modulus and Argument)

Let $z, w \in \mathbb{C}$.

1. $|zw| = |z||w|$
2. $\left|\dfrac{z}{w}\right| = \dfrac{|z|}{|w|}$
3. *An argument (not necessarily principal) of zw is* $\mathrm{Arg}(z) + \mathrm{Arg}(w)$.
4. *An argument (not necessarily principal) of* $\dfrac{z}{w}$ *is* $\mathrm{Arg}(z) - \mathrm{Arg}(w)$.

- Theorem 2.22 makes calculating moduli and arguments easier.

Homework 12

Use Property 1 to prove Property 2.

Example 2.23

Using the above properties, evaluate

$$\left| \frac{-2(3-i)(5+2i)}{(1+3i)(7-i)} \right|$$

2.3 Polar Form: Modulus and Argument

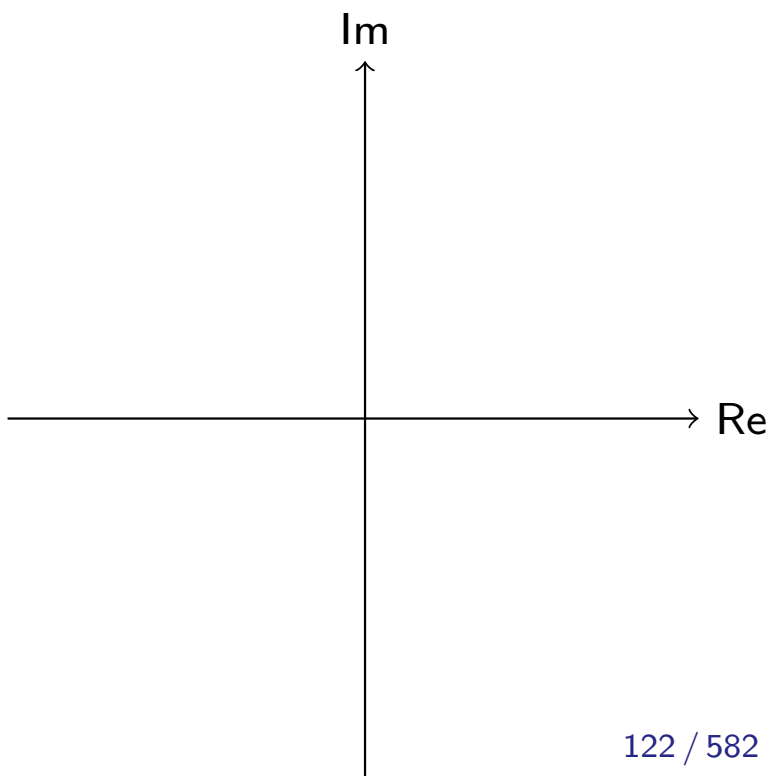
Example 2.24

Using the above properties, find an argument of

1. $z_1 = (1+i)(-1+\sqrt{3}i)$
2. $z_2 = \dfrac{-i}{-2+2i}$

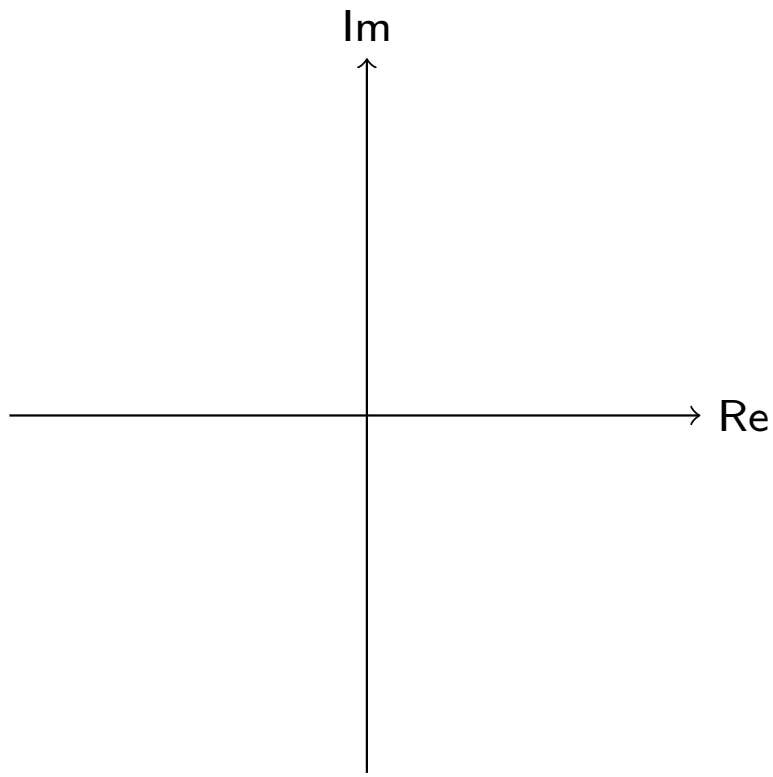
and hence find $\mathrm{Arg}(z_1)$ and $\mathrm{Arg}(z_2)$.

1.



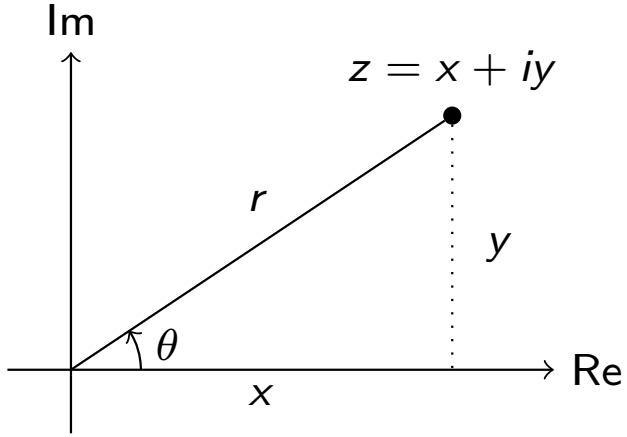
Example continued

2.



2.3 Polar Form: Trigonometric Polar Form

- Recall that any complex number $z = x + iy \neq 0$ can also be specified by its modulus r and an argument θ.



- From the diagram:

$$\cos(\theta) = \frac{x}{r} \text{ and } \sin(\theta) = \frac{y}{r} \Rightarrow x = r\cos(\theta) \text{ and } y = r\sin(\theta).$$

$$\Rightarrow z = x + iy = r\cos(\theta) + i\,r\sin(\theta).$$

Definition 2.25

Let $z \in \mathbb{C}$, with modulus r and argument θ. The <u>trigonometric polar form</u> of z is given by:

$$z = r(\cos(\theta) + i\sin(\theta)).$$

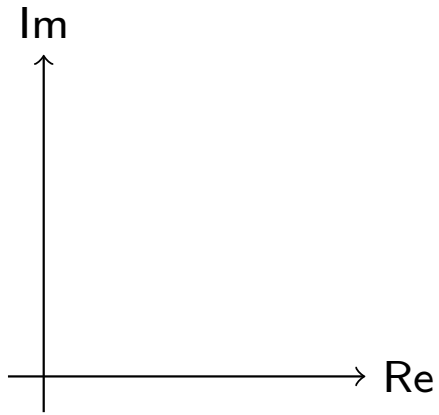
2.3 Polar Form: Trigonometric Polar Form

Example 2.26

Express the following complex numbers in trigonometric polar form.

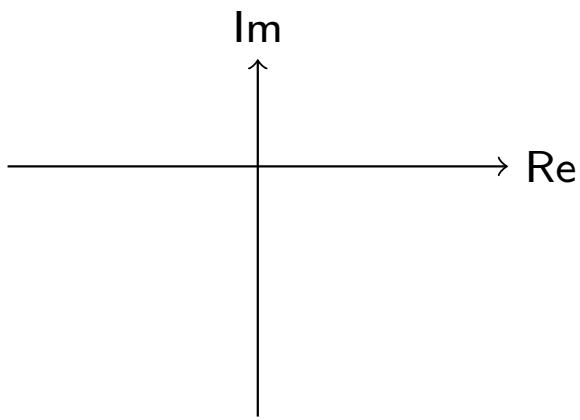
(a) $z = \sqrt{3} + i$ (b) $z = -1 - i$

(a)



2.3 Polar Form: Trigonometric Polar Form

(b)



2.3 Polar Form

Recall that our goal is to find a form for complex numbers that makes exponentiation easier. Unfortunately, trigonometric polar form doesn't quite do this.

$$(\cos \pi/2 + i \sin \pi/2)^{6000}$$

There is another polar form for complex numbers that simplifies calculations like this.

2.3 Polar Form: Exponential Polar Form

On every top 10 list of best mathematical equations, the following is usually number 1:

$$e^{i\pi} = -1$$

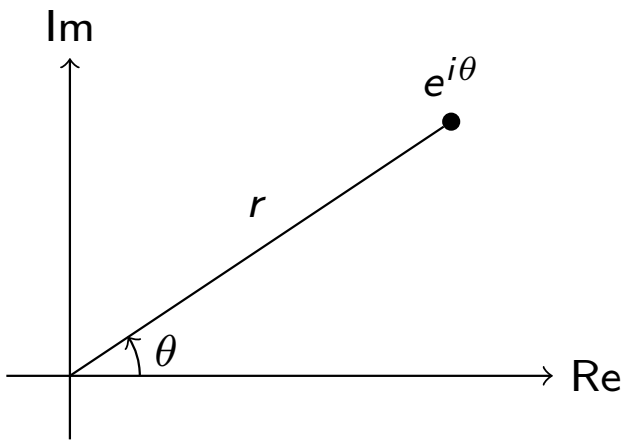
This looks like it has something to do with complex numbers: we see an i and an angle, π.

2.3 Polar Form: Exponential Polar Form

Definition 2.27

Let $z \in \mathbb{C}$, with modulus r and argument θ. The exponential polar form of z is given by:

$$z = re^{i\theta}$$



- This definition enables us to write a complex number $z \neq 0$ in exponential polar form

$$z = re^{i\theta}$$

where $r = |z|$ and θ is any argument of z.

2.3 Polar Form: Exponential Polar Form

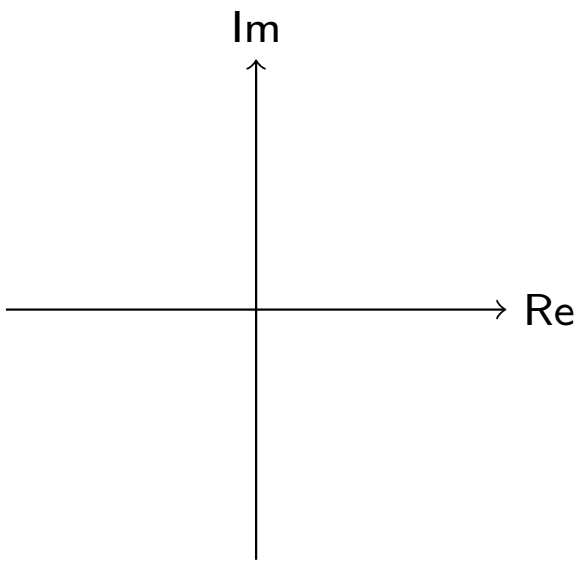
On every top 10 list of best mathematical equations, the following is usually number 1:

$$e^{i\pi} = -1$$

2.3 Polar Form: Exponential Polar Form

Example 2.28

Express $z = -2 + 2\sqrt{3}i$ in exponential polar form $re^{i\theta}$ and use it to compute z^2.



2.3 Polar Form: Exponential Polar Form

Example 2.29

Express $z = 5e^{i\frac{3\pi}{4}}$ in Cartesian form $x + iy$.

2.3 Polar Form: Exponential Polar Form

Homework 13

Express $z = 4e^{-i\frac{\pi}{3}}$ *in Cartesian form.*

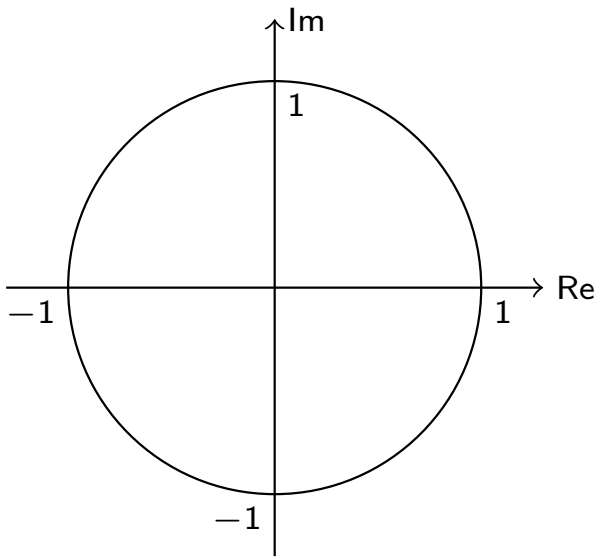
2.3 Polar Form: Exponentials and Conjugates

Example 2.30

Use the trigonometric identities $\cos(-\theta) = \cos(\theta)$ (even function) and $\sin(-\theta) = -\sin(\theta)$ (odd function) to prove

$$e^{-i\theta} = \overline{e^{i\theta}}$$

for any θ.



2.3 Polar Form: Complex Exponentiation

Theorem 2.31

1. $e^{i0} = 1$
2. $e^{i\alpha}e^{i\beta} = e^{i\alpha+i\beta} = e^{i(\alpha+\beta)}$
3. $\dfrac{e^{i\alpha}}{e^{i\beta}} = e^{i\alpha-i\beta} = e^{i(\alpha-\beta)}$
4. $e^{i\alpha} = e^{i\beta}$ *precisely if* $\beta = \alpha + 2k\pi$ *for some* $k \in \mathbb{Z}$.

- Properties 1–3 look like the usual rules for exponentials in $\mathbb{R}$
- This is why we generally work with the exponential polar form $re^{i\theta}$ instead of the trigonometric polar form $r(\cos\theta + i\sin\theta)$.

2.3 Polar Form: Complex Exponentiation

Property 2 in Theorem 2.31

- Recall the compound angle formulas for cos and sin:

$$\cos(\alpha+\beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$$
$$\sin(\alpha+\beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta).$$

Example 2.32

Use the compound angle formulas to prove Property 2.

2.3: Polar Form: Multiplication

Theorem 2.33

For two complex numbers $z_1 = re^{i\alpha}$ and $z_2 = se^{i\beta}$, the product is given by

$$z_1 z_2 = rse^{i(\alpha+\beta)}.$$

- For modulus, this gives the simple identity $|z_1 z_2| = |z_1||z_2|$.
- We can not write quite such a simple formula for the principal argument of a product.

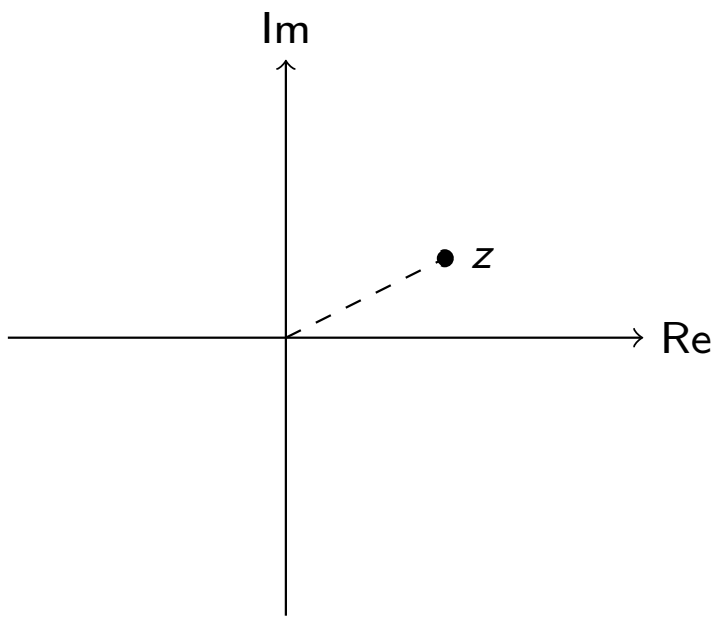
Homework 14

Find $z_1, z_2 \in \mathbb{C}$ such that $\mathrm{Arg}(z_1 z_2) \neq \mathrm{Arg}(z_1) + \mathrm{Arg}(z_2)$.

2.3 Polar Form: Multiplication by i

Example 2.34

Describe geometrically what happens when a complex number z is multiplied by $w = i$.



2.3 Polar Form: Multiplication by i

Homework 15

Describe geometrically what happens when a complex number z is multiplied by $w = -i$.

2.3 Polar Form: Division

Homework 16

Consider two complex numbers written in polar form:

$$z_1 = re^{i\alpha} \qquad \textit{and} \qquad z_2 = se^{i\beta} \neq 0.$$

Prove

$$\frac{z_1}{z_2} = \frac{r}{s}\, e^{i(\alpha-\beta)}.$$

1.5 Complex Numbers: Complex Division

- For modulus, this gives the simple identity $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$.
- We can not write quite such a simple identity for the principal argument of a quotient.

Homework 17

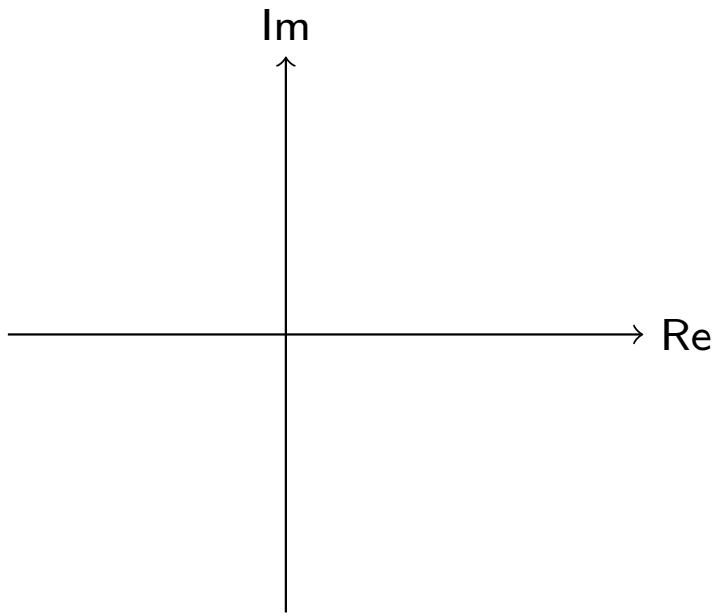
Find a pair of complex numbers of z_1 and z_2 such that

$$\operatorname{Arg}\left(\frac{z_1}{z_2}\right) \neq \operatorname{Arg}(z_1) - \operatorname{Arg}(z_2).$$

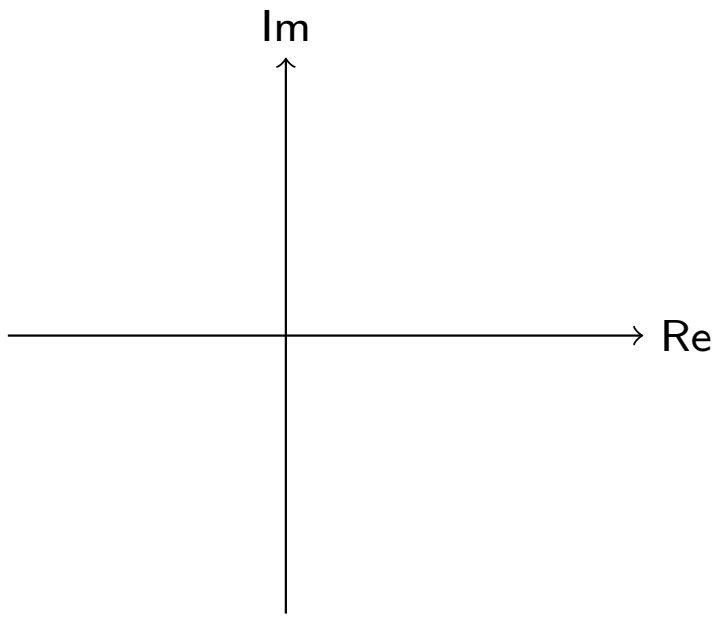
2.3 Polar Form: Complex Division

Example 2.35

Let $z = \sqrt{3} + i$ and $w = 1 - \sqrt{3}i$. Use exponential polar form to find $\frac{1}{z}$ and $\frac{z}{w}$.



2.3 Polar Form: Complex Division



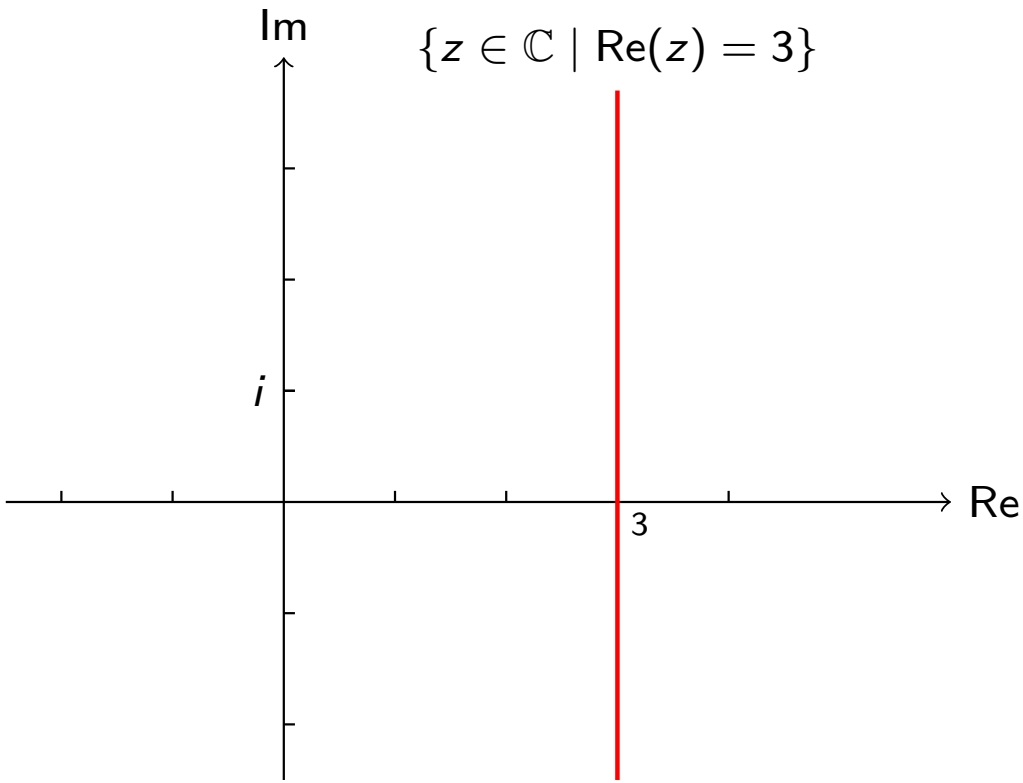
2.3 Polar Form: Complex Division

Homework 18

Describe geometrically what happens when a complex number z is divided by $w = \frac{1}{2} + \frac{\sqrt{3}}{2} i$.

2.4 Sketching subsets of the Complex Plane

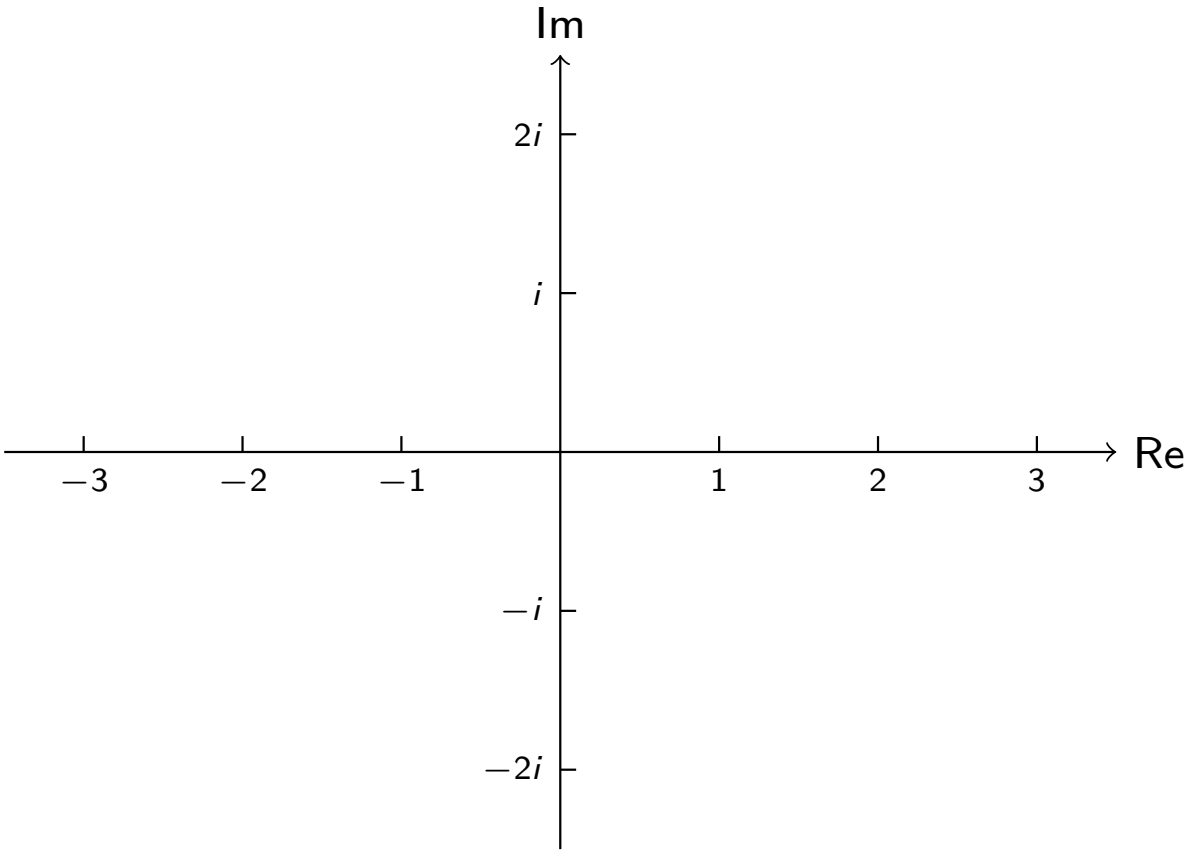
- The set of complex numbers that satisfy certain conditions corresponds to some region in the complex plane.
- **Example:** The set $\{z \in \mathbb{C} \mid \mathrm{Re}(z) = 3\}$ of all complex numbers with real part 3, is represented in the complex plane by the vertical line intersecting the real axis at 3.



2.4 Sketching subsets of the Complex Plane

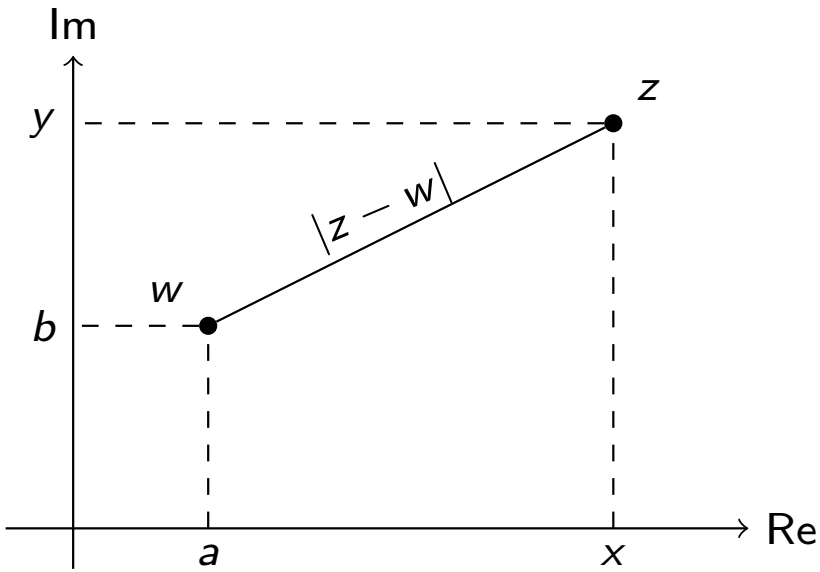
Example 2.36

Sketch the set $\{z \in \mathbb{C} \mid \mathrm{Im}(z) < 2\}$ in the complex plane.



2.4 Sketching subsets of the Complex Plane: Distance

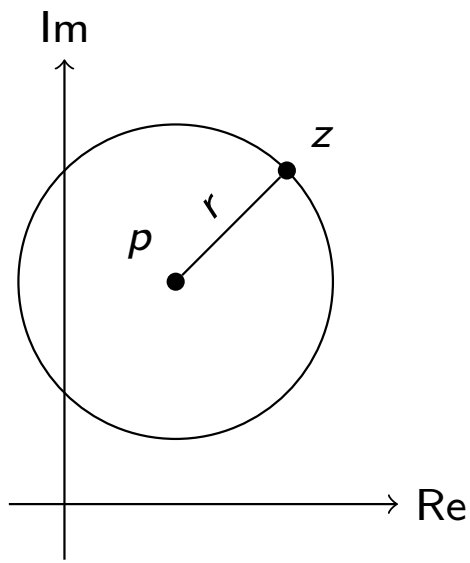
- For $z = x + iy$ and $w = a + ib$ in $\mathbb{C}$, the modulus $|z - w|$ is simply the distance between z and w in the complex plane.



- For fixed $p \in \mathbb{C}$ and $r > 0$ this means

$$C = \{z \in \mathbb{C} : |z - p| = r\}$$

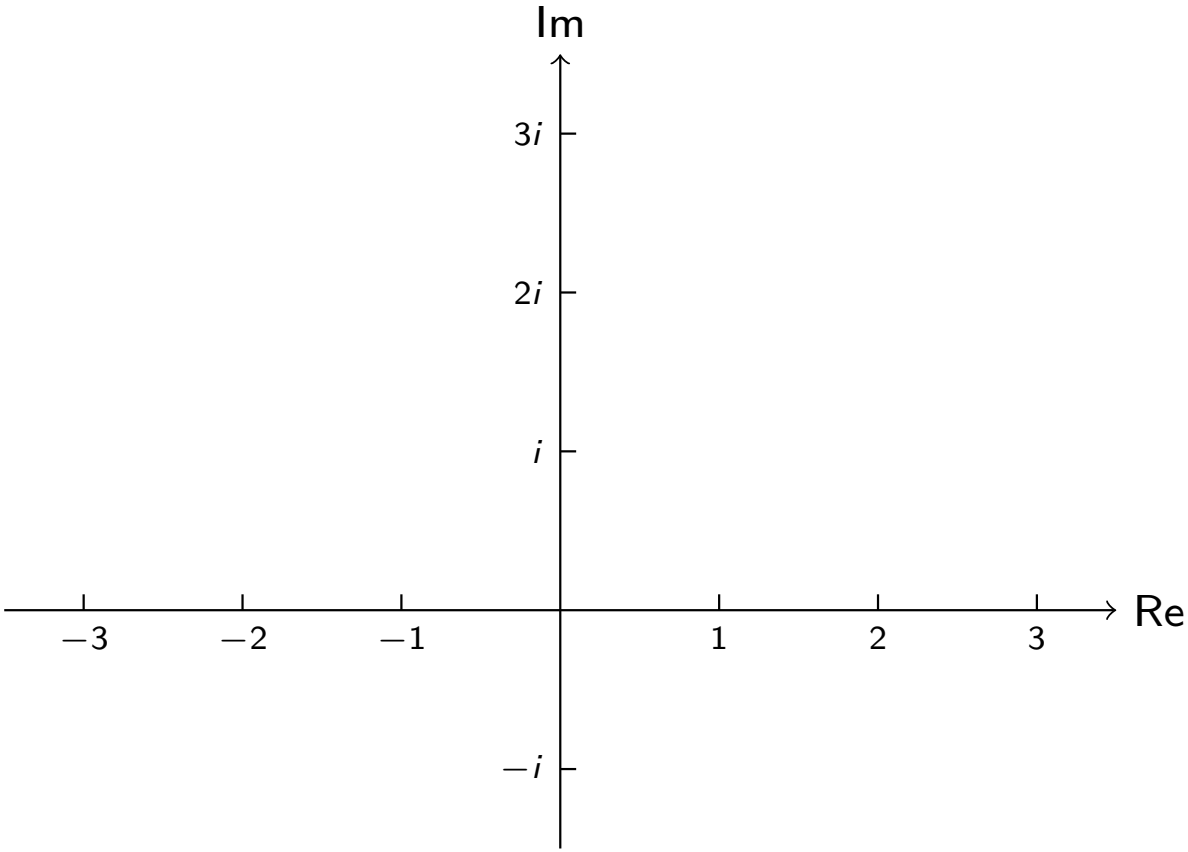
is a circle of radius r and centre p.



2.4 Sketching subsets of the Complex Plane

Example 2.37

Sketch the set $C = \{z \in \mathbb{C} : |z - i| = 2\}$.



2.4 Sketching subsets of the Complex Plane

Find the Cartesian equation (i.e. an equation in terms of x and y) of this curve, by substituting $z = x + iy$ into

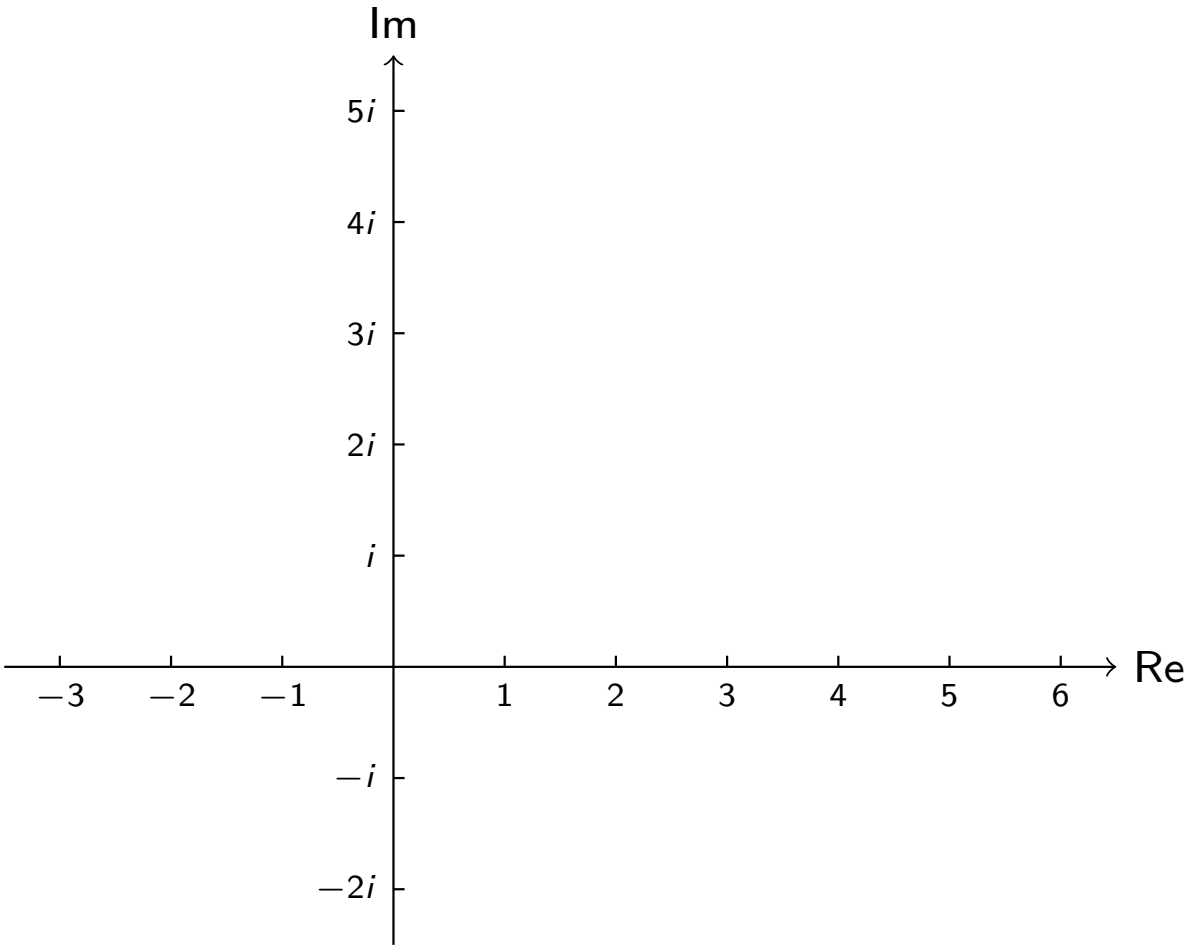
$$|z - i| = 2$$

and simplifying.

2.4 Sketching subsets of the Complex Plane

Example 2.38

Sketch the set $\{z \in \mathbb{C} : |z - 3 - 2i| \leq 3\}$.



2.4 Sketching subsets of the Complex Plane

Homework 19

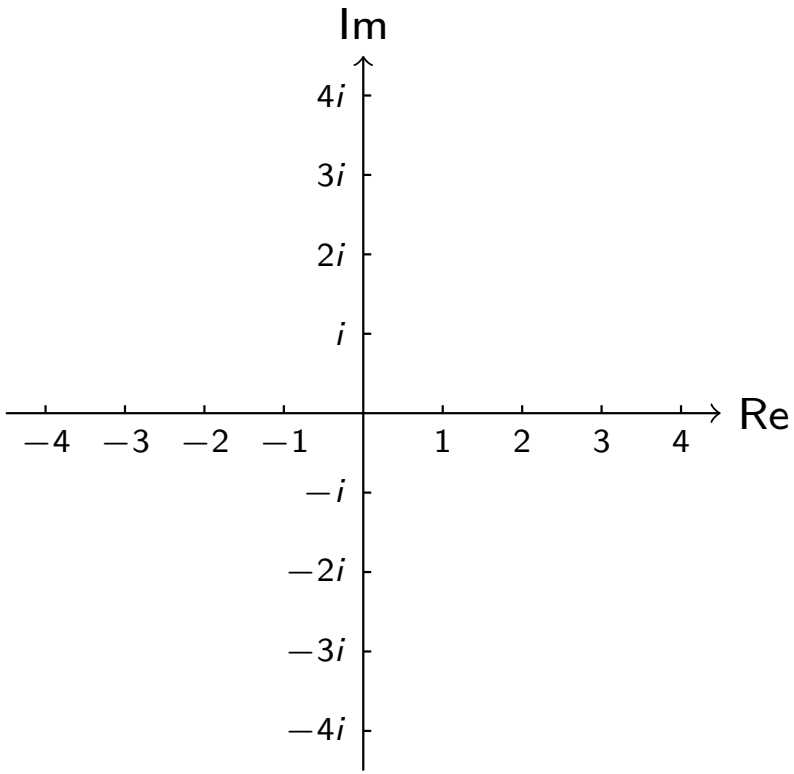
Find the Cartesian equation of this region by substituting $z = x + iy$. Hence find the points where the boundary of the set in Example 2.38 intersects the real axis.

2.4 Sketching subsets of the Complex Plane

Example 2.39

Sketch the set

$$A = \left\{ z \in \mathbb{C} \;\middle|\; 2 \leq |z| \leq 4 \quad \text{and} \quad \frac{\pi}{4} < \operatorname{Arg}(z) < \frac{3\pi}{4} \right\}.$$



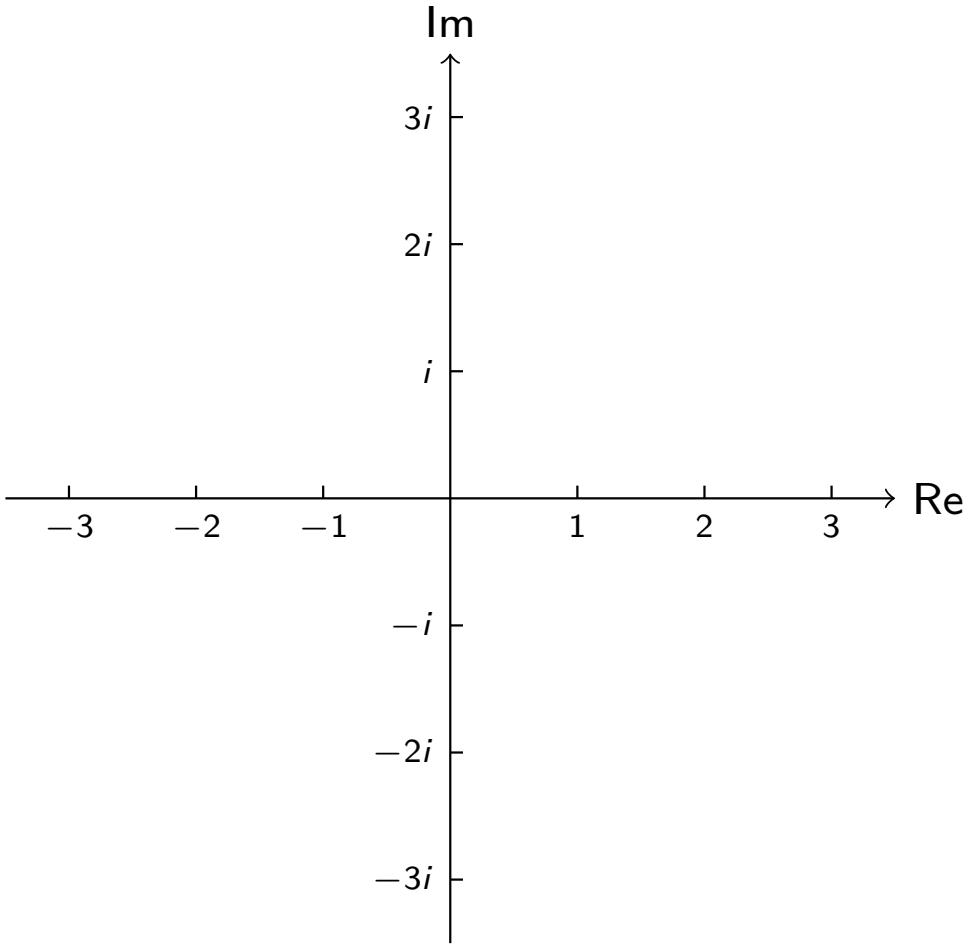
2.4 Sketching subsets of the Complex Plane

Example 2.40

Use a geometric argument based on the fact that $|z - a|$ gives the distance between z and a to sketch the set

$$B = \{z \in \mathbb{C} : |z + 2| = |z + i|\}.$$

2.4 Sketching subsets of the Complex Plane



2.4 Sketching subsets of the Complex Plane

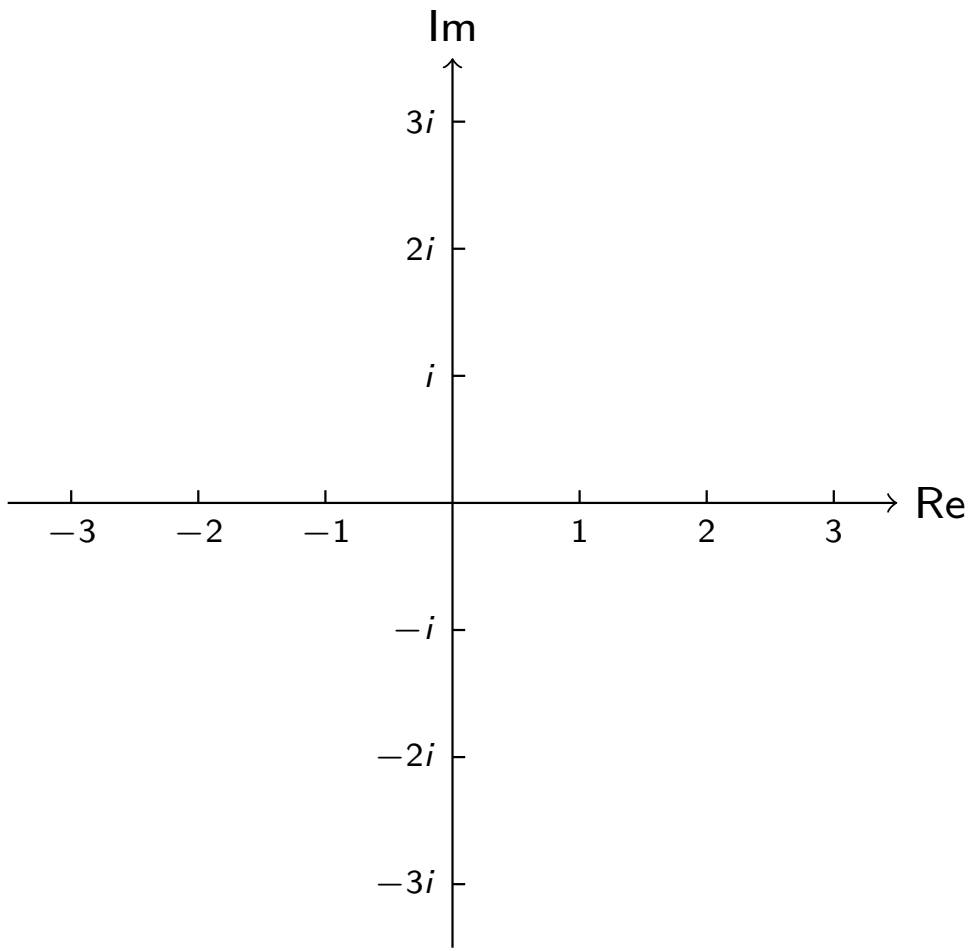
Homework 20

Find the Cartesian equation of the set in Example 2.40.

2.4 Sketching subsets of the Complex Plane

Example 2.41

By expressing a typical element of $\mathbb{C}$ in Cartesian form, determine the set $\{z \in \mathbb{C} \mid z = i\bar{z}\}$ and sketch it.



2.5 Powers and Roots of Complex Numbers

How can we find

$$(\cos(1) + i\sin(1))^{100}?$$

2.5 Powers and Roots of Complex Numbers: De Moivre's Theorem

Theorem 2.42 (De Moivre's theorem)

Let $z = re^{i\theta}$. For any integer n:

$$z^n = r^n e^{in\theta}.$$

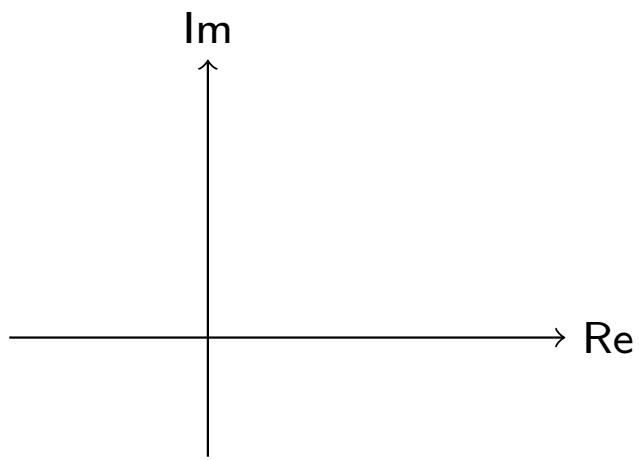
Proof idea:

2.5 Powers and Roots of Complex Numbers: De Moivre's Theorem

De Moivre's theorem can be used to avoid expanding the brackets when finding large powers of complex numbers.

Example 2.43

Use de Moivre's theorem to find $(1 + i\sqrt{3})^8$ in both exponential and Cartesian form.

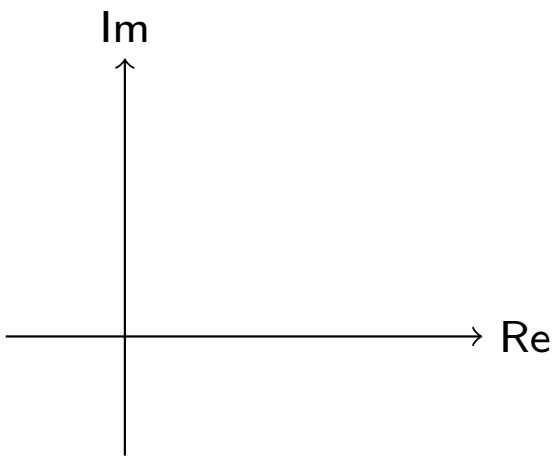


Example continued

2.5 Powers and Roots of Complex Numbers: De Moivre's Theorem

Example 2.44

Find $\left(\dfrac{2}{1+i}\right)^{14}$ in exponential and Cartesian form.



Example continued

2.5 Powers and Roots of Complex Numbers: Roots

- Suppose we wish to find all n-th roots of a fixed complex number $w = se^{i\alpha} \neq 0$. This means finding all solutions z of:

$$z^n = w.$$

2.5 Powers and Roots of Complex Numbers: Roots

- The values $k = 0, 1, \ldots, n-1$ give n distinct n-th roots of w.
- But we can stop at $n-1$ since:
  - Plugging in $k = n$ gives the same z-value as $k = 0$.
  - Plugging in $k = n+1$ gives the same answer as $k = 1$.
  - Plugging in $k = n+2$ gives the same answer as $k = 2$.
  - And so on ...

Example 2.45

Check the above claims in the cases $k = 0$ and $k = 1$.

2.5 Powers and Roots of Complex Numbers: Roots

- The reasoning on the two previous slides amounts to a proof of the following theorem.

Theorem 2.46 (Root finding formula)

Every $w = se^{i\alpha} \in \mathbb{C} \setminus \{0\}$ *has exactly* n *complex* n*-th roots given by:*

$$w^{\frac{1}{n}} = s^{\frac{1}{n}}\, e^{i(\frac{1}{n}(\alpha+2k\pi))}$$

for $k = 0, 1, \ldots, n-1$. *In set notation:*

$$\{z \in \mathbb{C} \mid z^n = w\} = \left\{ s^{\frac{1}{n}}\, e^{i(\frac{1}{n}(\alpha+2k\pi))} \mid \; k = 0, 1, \ldots, n-1 \right\}$$

2.5 Powers and Roots of Complex Numbers: Roots

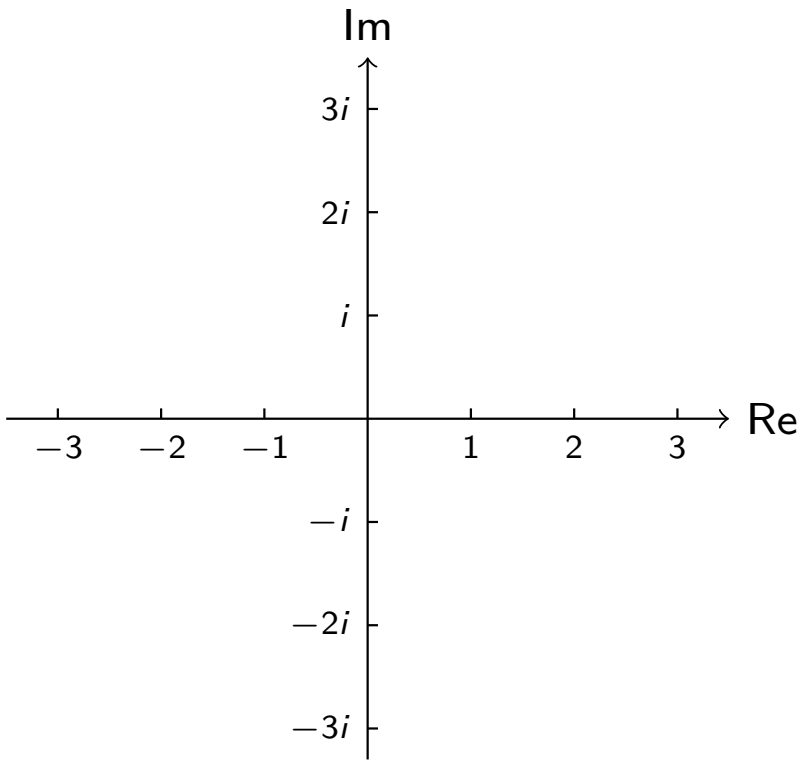
Example 2.47

Find the set of cube roots of -8.

- Over the reals, -2 is the only cube root of -8.
- Over the complex numbers -8 has 3 cube roots!
- Note: Our root finding formula does not necessarily express angles as principal arguments.

2.5 Powers and Roots of Complex Numbers: Roots

Sketch these 3 cube roots of -8 and check that they are correct:



- Observe that the cube roots of -8 are evenly spaced around a circle of radius 2.
- In general, the n-th roots of a complex number are always evenly spaced around a circle centred at the origin.

2.5 Powers and Roots of Complex Numbers: Roots

Express the three cube roots of -8 in Cartesian form:

2.5 Powers and Roots of Complex Numbers: Roots

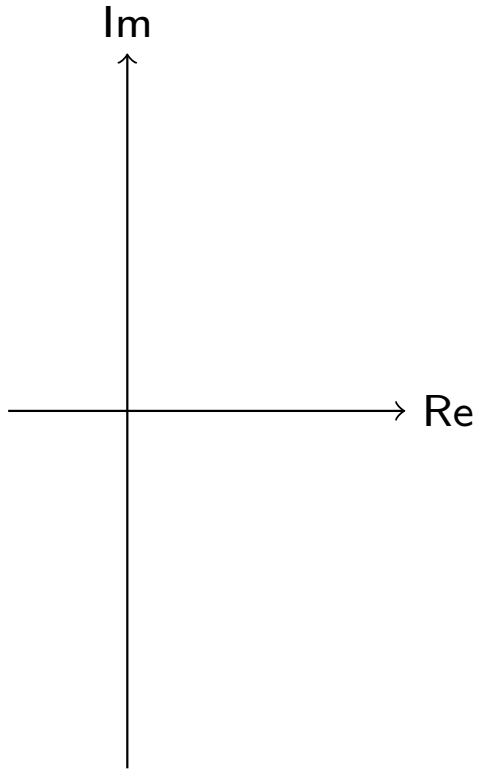
Homework 21

Find the 6*-th roots of* 1*, and sketch these in the complex plane.*
You may keep your answers in exponential polar form.
Note: *The* n*-th roots of* 1 *are often called 'roots of unity'.*

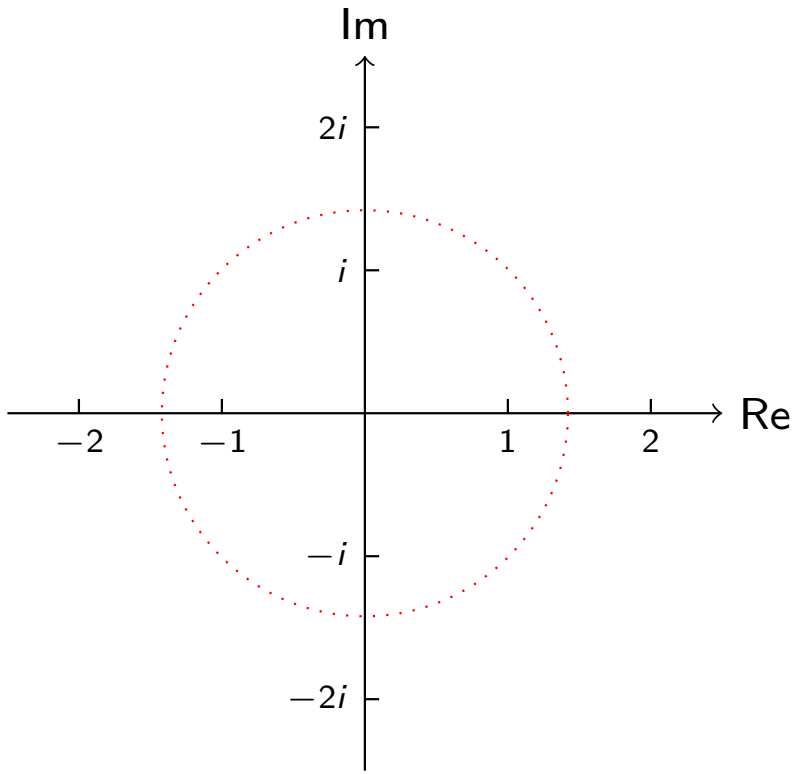
2.5 Powers and Roots of Complex Numbers: Roots

Example 2.48

Find the 4-th roots of $2 - 2i\sqrt{3}$ and sketch them.



2.5 Powers and Roots of Complex Numbers: Roots



2.5 Powers and Roots of Complex Numbers: Roots

Homework 22

Find the cube roots of $1 + i$.

2.6 Complex Representation of Trigonometric Functions

- The complex exponential can be used to express cos and sin in a way that makes a lot of calculations easier.

Example 2.49

Use Theorem 2.14(Part 1) and Example 2.30 to prove that

$$\cos(\theta) = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)$$

for every $\theta \in \mathbb{R}$.

2.6 Complex Representation of Trigonometric Functions

Homework 23

Prove $\sin(\theta) = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)$.

Theorem 2.50

$\cos(\theta) = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right)$ *and* $\sin(\theta) = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)$.

- We can use these identities to express powers of $\cos(\theta)$ or $\sin(\theta)$ in terms of $\cos(n\theta)$ or $\sin(n\theta)$.
- This will be very useful in our study of integration.

2.6 Complex Representation of Trigonometric Functions

- Applying the binomial formula to our exponential formulas for sin and cos converts expressions $\sin^m(x)$ and $\cos^n(x)$ into a form where we can integrate them.

Example 2.51

Express $\sin^4(x)$ as a sum of sines or cosines of multiples of x.

Example continued

This approach always converts expressions of the form $\sin^m(x)\cos^n(x)$ to an equivalent form that is easy to integrate.

2.6 Complex Representation of Trigonometric Functions

Homework 24

Express $\cos^4(x)\sin^2(x)$ *as a sum of sines and/or cosines of multiples of* x.

2.7 Factoring Polynomials

We can apply our knowledge of complex numbers to the well known problem of factoring polynomials. Our method for finding roots of complex numbers supports new techniques for finding roots of polynomials and hence factoring them. An ability to factorise over the complex numbers often enables us to factor in the more familiar setting of real valued polynomials.

This has many applications in calculus. For example, we later learn how to integrate functions of the form

$$\frac{p(x)}{q(x)}$$

where $p(x)$ and $q(x)$ are polynomials using the technique of partial fractions. A key step in this technique is factoring of $q(x)$.

2.7 Factoring Polynomials: Complex Polynomials

- A *real polynomial* like

$$p(x) = 3x^4 + 5x^2 - x + 2$$

  extends to a *complex polynomial*

$$q(z) = 3z^4 + 5z^2 - z + 2$$

  defined for all $z \in \mathbb{C}$.
- Since these two polynomials agree exactly on $\mathbb{R}$, solving $q(z) = 0$, also yields all solutions of $p(x) = 0$.

2.7 Factoring Polynomials: Roots of Complex Polynomials

- Finding the roots of a complex polynomial $P(z)$ means (as usual) solving $P(z) = 0$.
- Recall that over the reals $\mathbb{R}$, some quadratics have roots precisely if they can be factorised into linear factors. E.g.,

$$x^2 + x - 2 = (x + 2)(x - 1),$$

  since this equation has roots -2 and 1.
- Others have no real roots and cannot be factored over $\mathbb{R}$. EG:

$$x^2 - 2x + 2.$$

  (Here the discriminant $b^2 - 4ac = -4$ is negative.)

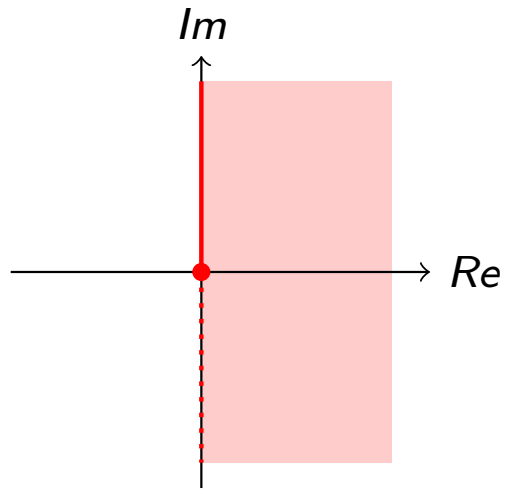
2.7 Factoring Polynomials: Roots of Complex Polynomials

- For $w = re^{i\theta} \in \mathbb{C} \setminus \{0\}$ where $\theta = \text{Arg}(w)$, the root finding formula gives two solutions of $z^2 = w$:

$$z_1 = \sqrt{r}e^{i(\frac{\theta}{2})}, \qquad z_2 = \sqrt{r}e^{i(\frac{\theta}{2}+\pi)} = -z_1$$

- z_1 is called the principal square root of w (sometimes denoted $\sqrt{w}$), so the square roots of w are $\pm\sqrt{w}$.

- Since $\theta = \text{Arg}(w) \Rightarrow \dfrac{\theta}{2} \in (-\dfrac{\pi}{2}, \dfrac{\pi}{2}]$, $\sqrt{w}$ is always in quadrant 1 or 4.



- For $w \in [0, \infty)$, this agrees with our usual definition of $\sqrt{w}$.
- For $w \in (-\infty, 0)$, $\sqrt{w}$ is on the positive imaginary axis.

2.7 Factoring Polynomials: The Quadratic Formula

- Since the algebra used to prove the quadratic formula holds in $\mathbb{C}$, we can still use the quadratic formula.
- This always gives linear factors.

Example 2.52

Use the quadratic formula to factorise $p(z) = z^2 - 3iz - 2$ over $\mathbb{C}$.

2.7 Factoring Polynomials: Factoring Complex Polynomials

- We have now achieved our aim of constructing a number system in which all polynomial equations have solutions.

Theorem 2.53 (The Fundamental Theorem of Algebra)

Every complex polynomial $P(z)$ of degree n can be factored into n linear factors over $\mathbb{C}$, that is

$$P(z) = a(z - \alpha_1)(z - \alpha_2)\dots(z - \alpha_n),$$

where $a, \alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{C}$.

- This also tells us that a polynomial of degree n has $\leq n$ roots over $\mathbb{C}$.
- There might be some repeated roots and hence repeated factors, so the number of distinc roots could be $< n$.

2.7 Factoring Polynomials: Conjugate Pairs

Theorem 2.54 (Conjugate Pairs)

If the coefficients of a complex polynomial $P(z)$ are all in $\mathbb{R}$, the non-real roots of $P(z)$ occur in complex conjugate pairs z and $\bar{z}$.

- Another way of stating this result is to say that:

$$P(z) = 0 \Rightarrow P(\bar{z}) = 0.$$

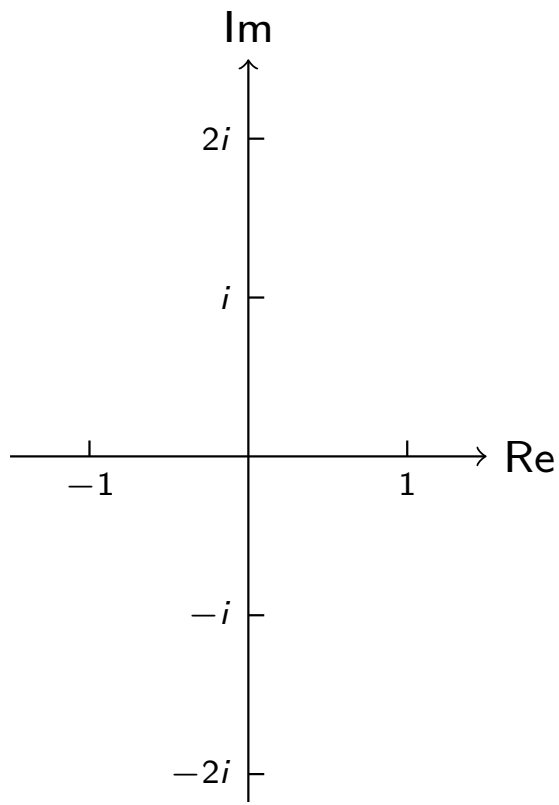
- The requirement that all coefficients are all real is indispensable, as our next example illustrates.

2.7 Factoring Polynomials: Conjugate Pairs

Example 2.55

Consider the polynomial $P(z) = z^3 - 3iz^2 - 2z$.

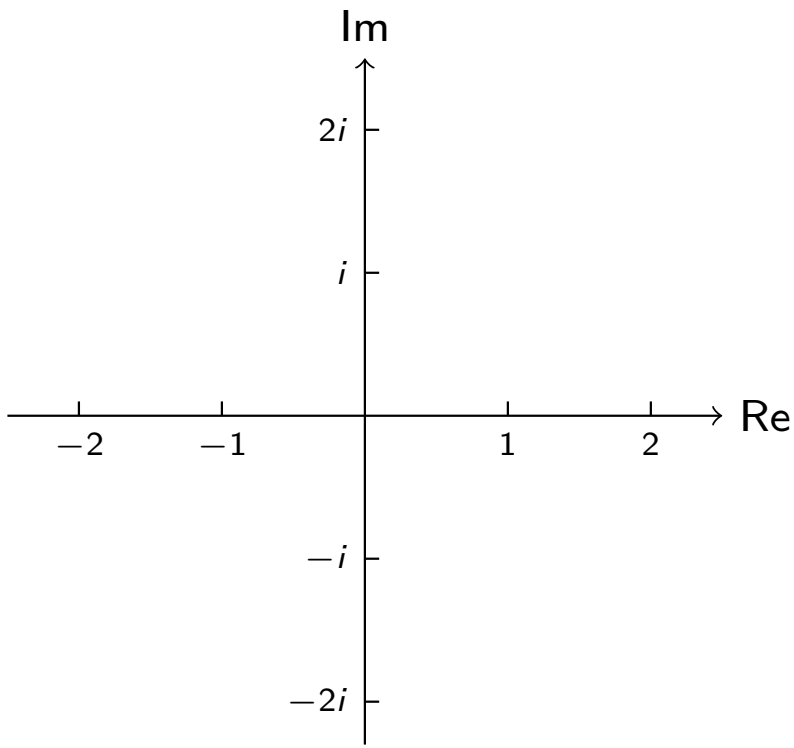
1. Factor $P(z)$.
2. Sketch the roots of $P(z)$.
3. Do the non-real roots come in conjugate pairs?



2.7 Factoring Polynomials: Factoring Complex Polynomials

Example 2.56

Sketch the set $\{z \in \mathbb{C} \mid \ z^4 + z^2 - 12 = 0\}$.



2.7 Factoring Polynomials

Homework 25

Sketch the set $\{z \in \mathbb{C} \mid \ z^4 - 2z^2 + 4 = 0\}$.

2.7 Factoring Polynomials: Conjugate Pairs

- The following consequence of the Conjugate Pairs Theorem has important consequences for our study of integration.
- A quadratic is called <u>irreducible</u> (over $\mathbb{R}$) when it has no real roots (and hence no real linear factors).

Theorem 2.57 (Real Polynomials)

If the coefficients of a polynomial $p(x)$ are all real, then $p(x)$ can be written as a product of linear and irreducible quadratic factors with real coefficients.

2.7 Factoring Polynomials: Conjugate Pairs

Example 2.58

Express the polynomial $p(z) = z^4 + z^2 - 12$ as a product of linear and quadratic factors with all coefficients real .

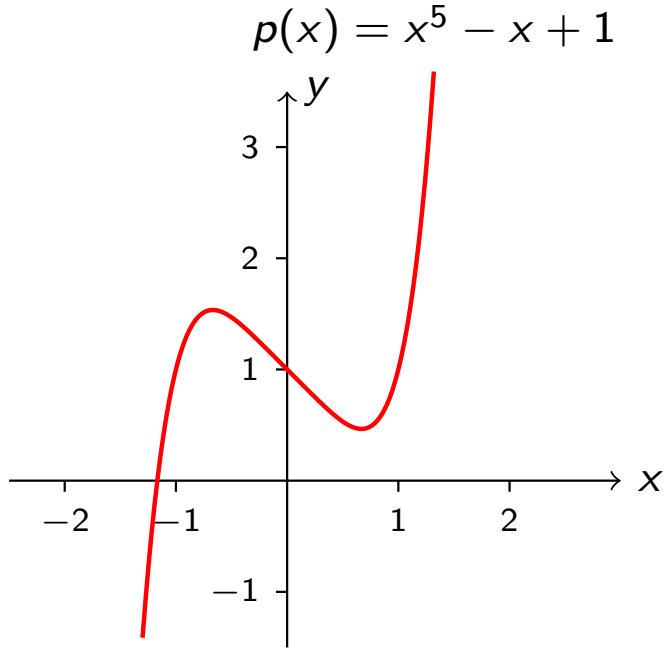
2.7 Factoring Polynomials: Conjugate Pairs

Homework 26

Express the polynomial $p(z) = (z-3)^3 + 8$ *as a product of linear and quadratic factors with real coefficients.*

2.7 Factoring Polynomials: Factoring Complex Polynomials

- Even though the Real Polynomials theorem guarantees linear/quadratic factors exist, they can be very difficult to find.
- For polynomials of degree 5 or more, it is often impossible!
- For example, we can not express the factors of $p(x) = x^5 - x + 1$ in simple terms, even though $p(x)$ has a real root.



- These results belong a field of mathematics called Galois Theory, studied in *MAST30005 Algebra*.